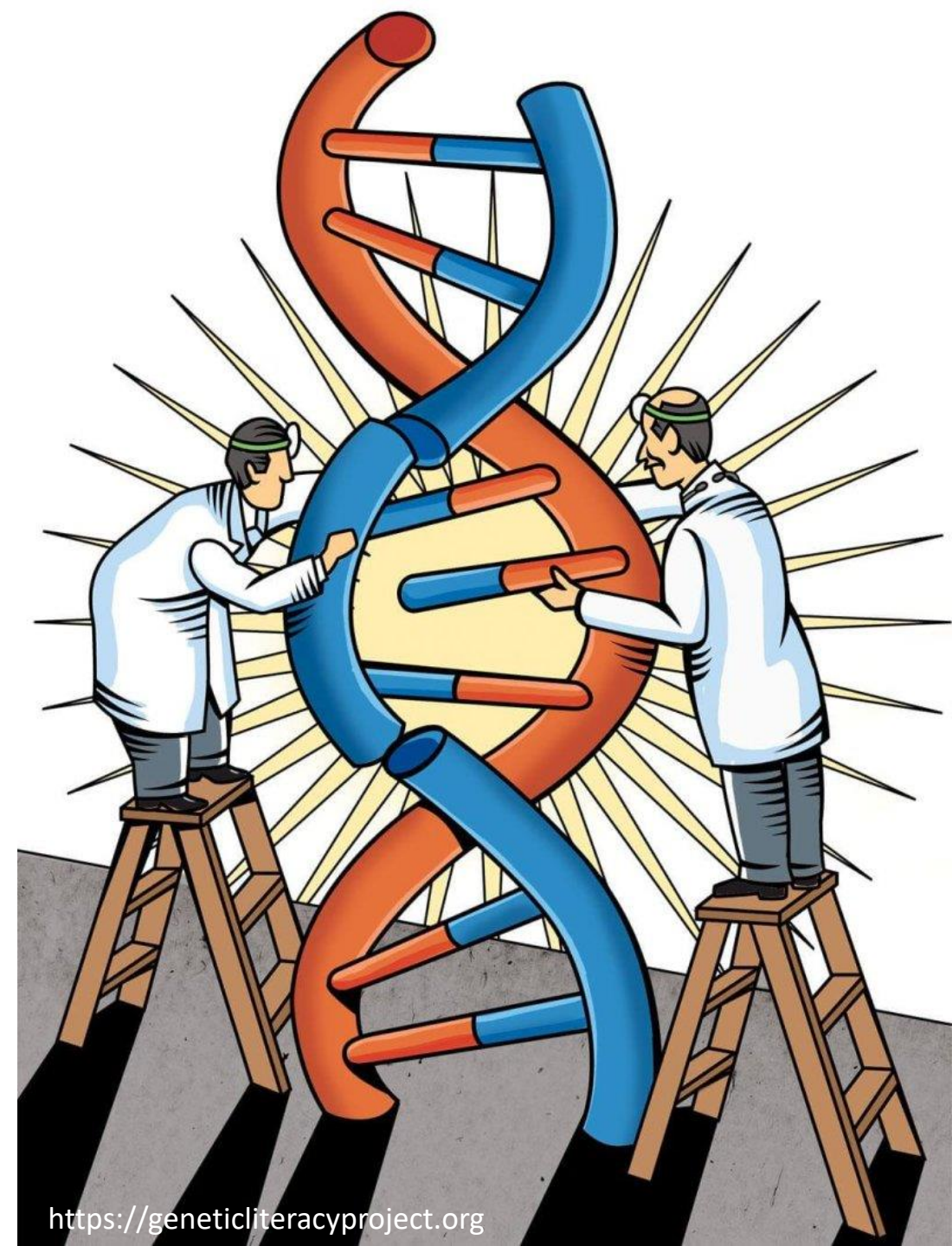


CRISPR in action

practice and theory of genome engineering

Day 2
August 18, 2026

Sibylle Vonesch
Laboratory for Genome Editing and
Systems Genetics



Why do I need different CRISPR vectors for different species?

Regulatory elements

- Promoters/terminators differ across species, different architecture pro- vs eukaryotes
- Cas9 and gRNA often need strong constitutive or inducible promoters and efficient terminators that are species-optimized

Codon usage

- Cas9 ORF often needs codon optimization to match host translation preferences

Selection markers

- Antibiotic or auxotrophy markers must match (e.g. *URA3* in yeast vs. hygromycin in mammalian cells) the organism's selectable traits and resistance background.

Origins of replication & plasmid maintenance

- Plasmid origins are not universal
- Stability and copy number depend on species-specific replication/segregation systems

Other elements

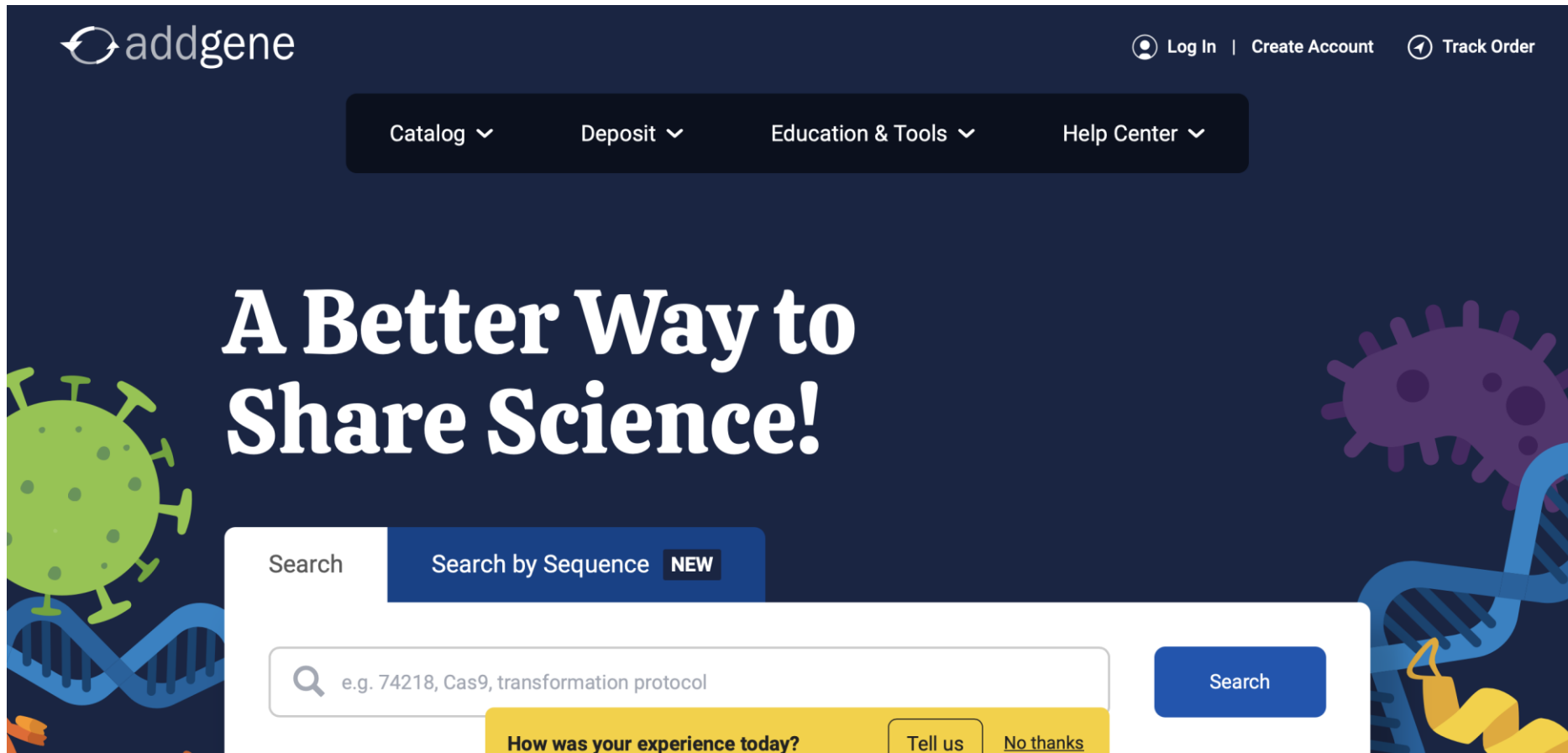
- Nuclear localization signals (NLS) must match host nuclear import machinery.

Take-home:

No single universal CRISPR plasmid works everywhere - each host requires tailored promoters, markers, and replication elements for stable and efficient gRNA & Cas9 expression.

Which CRISPR vector do I use for my species?

<https://www.addgene.org>



Optimal gRNA expression

High expression = better efficacy

Bacteria: strong constitutive promoters or inducible (Tet, Lac, IPTG)

Eukaryotes:

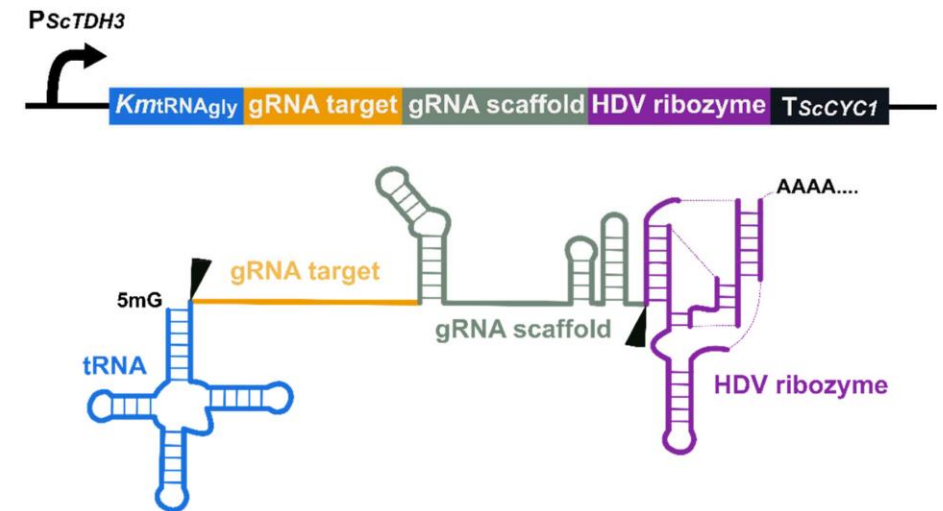
Pol III promoters (tRNA, snoRNA, snRNA (U6))

- high expression levels
- no processing
- More sensitive to poly T (some differences)

Pol II promoters

- High expression levels
- Inducible, tissue-specific expression
- Ribozymes or other processing elements

Day 4 for more specific recommendations in mammalian cells!



<https://doi.org/10.1016/j.mec.2022.e00208>

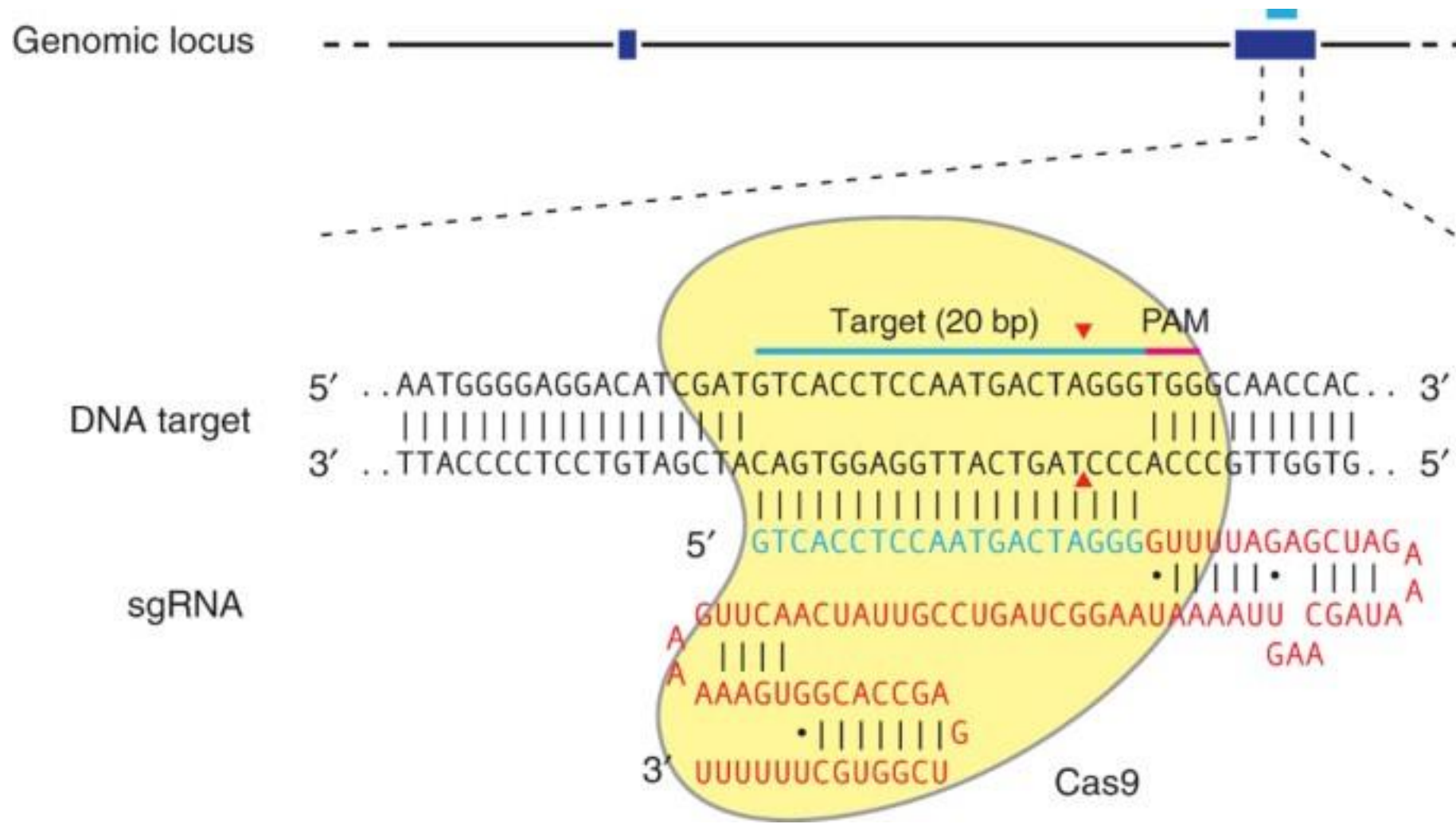
Optimal Cas expression

- Bacteria:
 - low to moderate or inducible expression lowers cellular toxicity, recleavage (SNPs)
 - typically on plasmid for rapid curing
- Eukaryotes:
 - Typically moderate to strong constitutive or inducible/tissue-specific promoters
 - Plasmids, viral delivery, integrative constructs
 - In mammalian cells often temporary expression (minimizes off-targets) - delivery as *in vitro* transcribed mRNA, Ribonucleoprotein (RNP) complexes (Cas9 protein + *in vitro* transcribed gRNA)

Day 4 for more specific recommendations in mammalian cells!

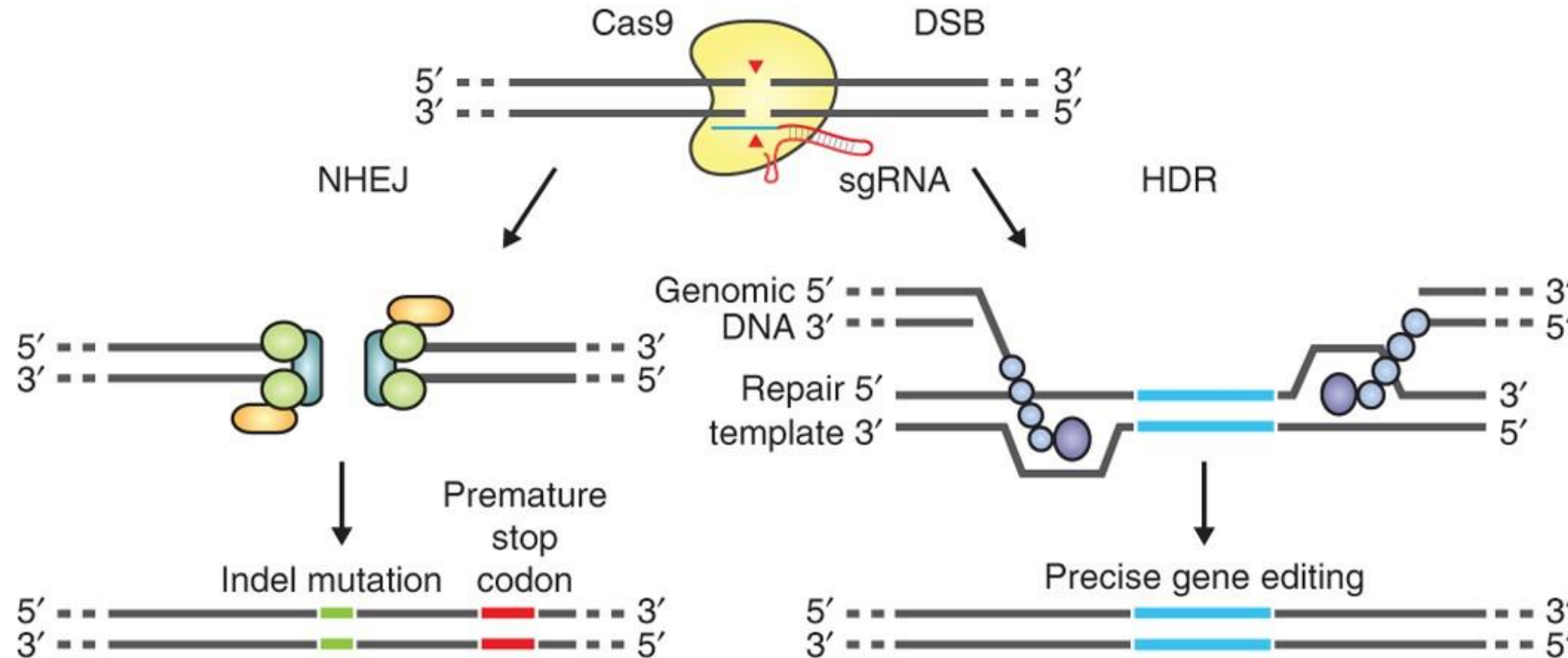
Repair pathways

Components of the *S. pyogenes* CRISPR/Cas9 system



Genome modification using CRISPR/Cas

Double-strand break is lethal → repair by endogenous host pathways is required

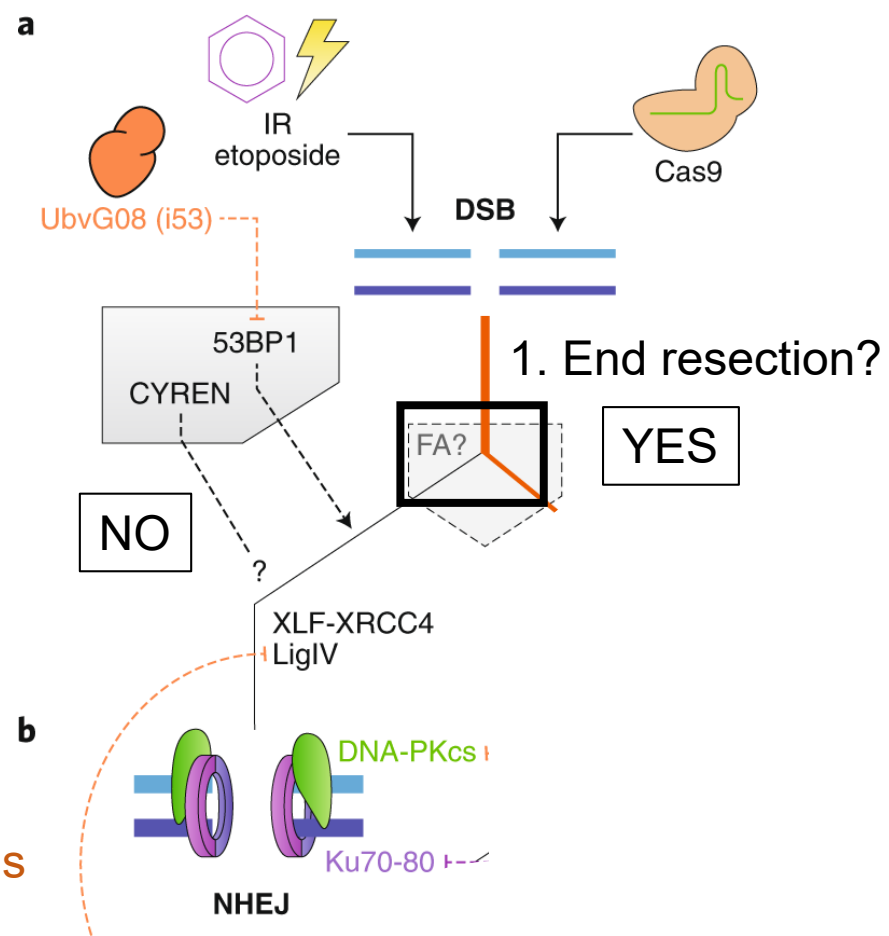


Non-Homologous End Joining:

Indels → frameshift → premature stop
→ present in eukaryotes and only a few prokaryotes

Homology-Directed Repair

→ recombination with intact, homologous rescue DNA (= donor), introduction of designed sequences



- **Key NHEJ vs HDR decision point: end resection**

NHEJ outcomes

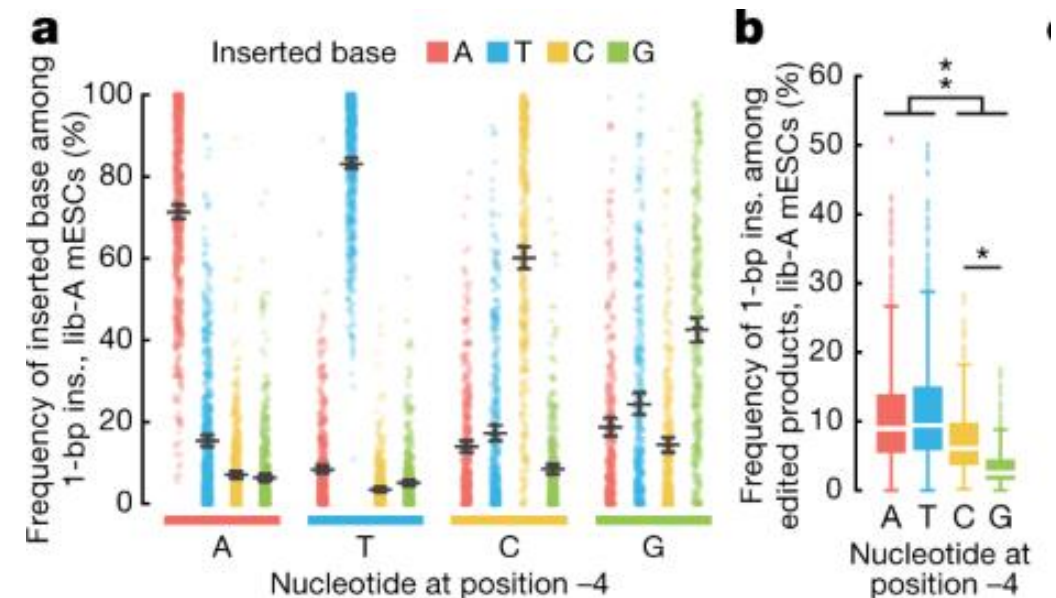
Generally heterogeneous, some loci more predictable

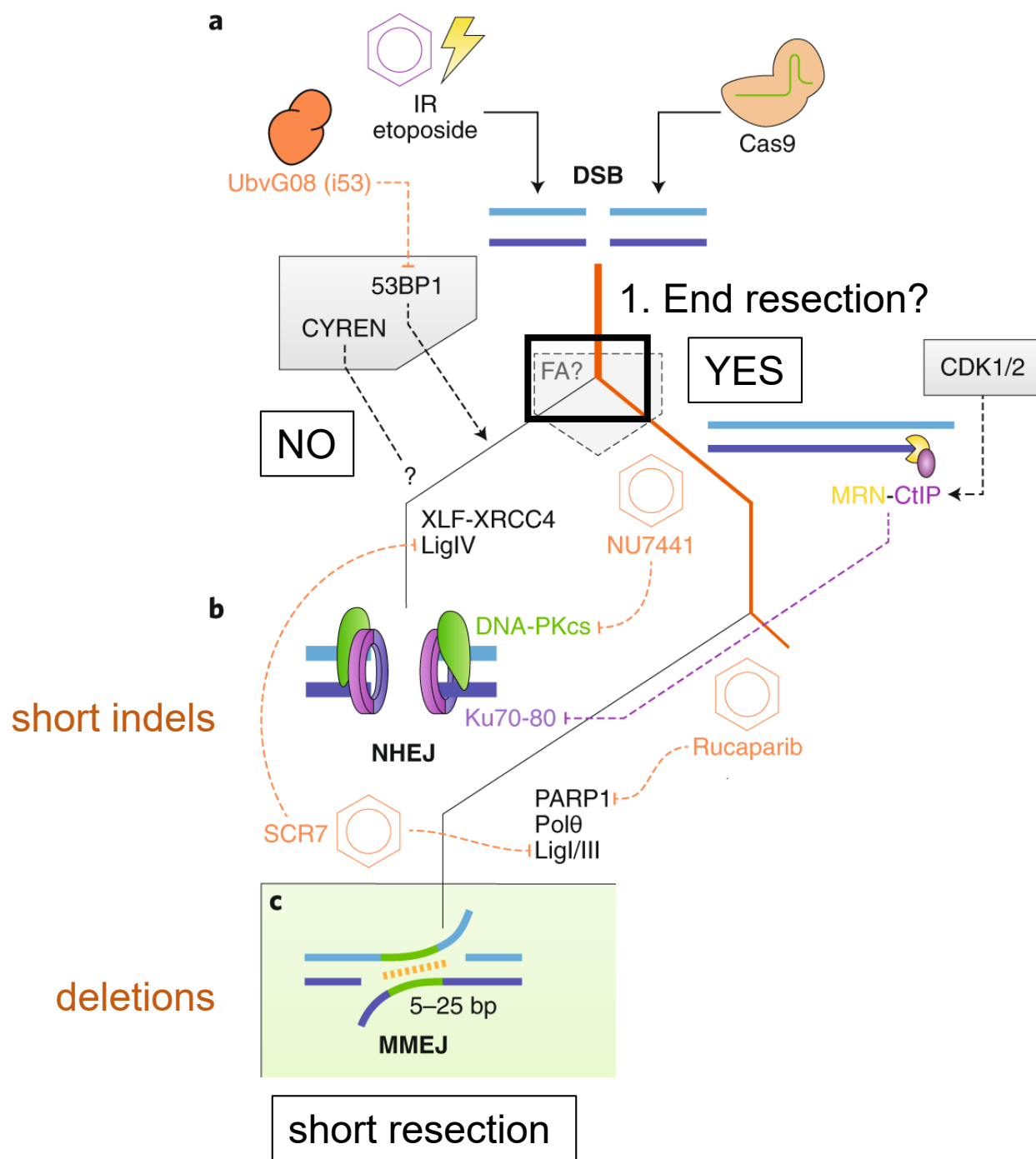
+1 insertion bias:

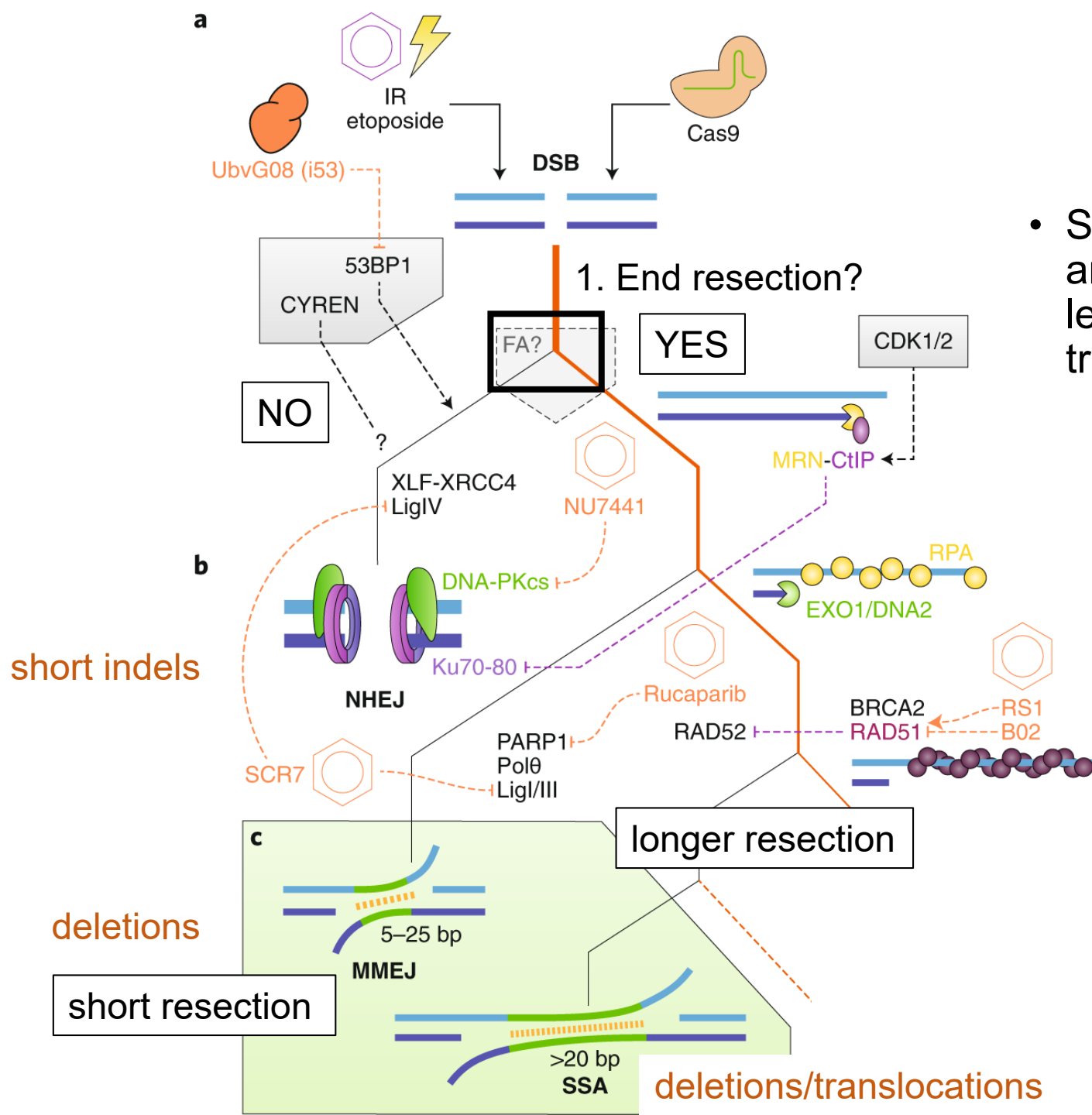
- most common NHEJ indel: **+1 insertion**.
- templated by the nucleotide at the -1 position
- Stronger bias if -1 is A or T
- Weaker if -1 is G or C (lower insertion frequency, deletions)
- +1 base can modulate effect
- Fill-in synthesis: staggered Cas9 cuts can leave 1-nt overhangs → polymerase fills in by copying -1 base
- Alignment/slippage: DNA polymerase misaligns during repair synthesis, duplicating base at -1

Selected data, endogenous HEK293 Obs.% |

CCGTCCAAGACCTA CCGAGGAGCTTTCC	
CCGTCCAAGACCTAACC GAGGAGCTTTCC	30.9
CCGTCCAAGACCT- CCGAGGAGCTTTCC	7.3
CCGTCCAAGACCTA -CGAGGAGCTTTCC	
CCGTCCAAGACC-- --GAGGAGCTTTCC	23.6
CCGTCC----- --GAGGAGCTTTCC	3.3
CCG----- --AGGAGCTTTCC	1.8
⋮	⋮

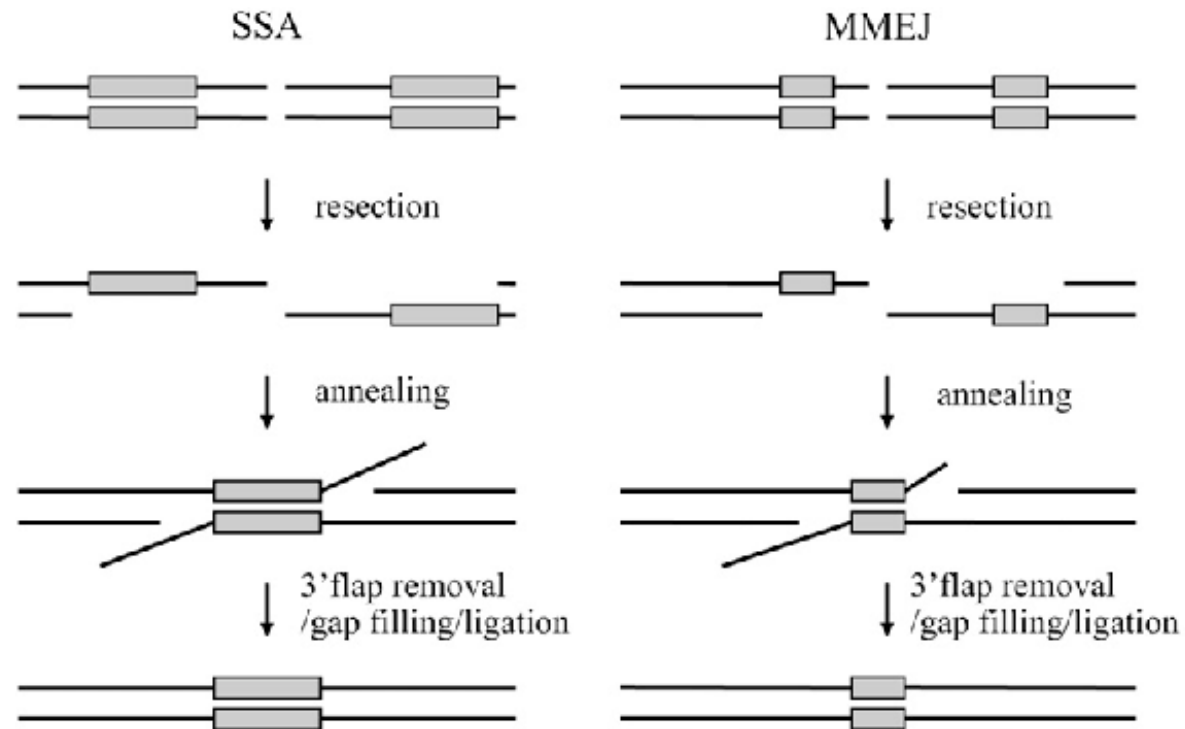


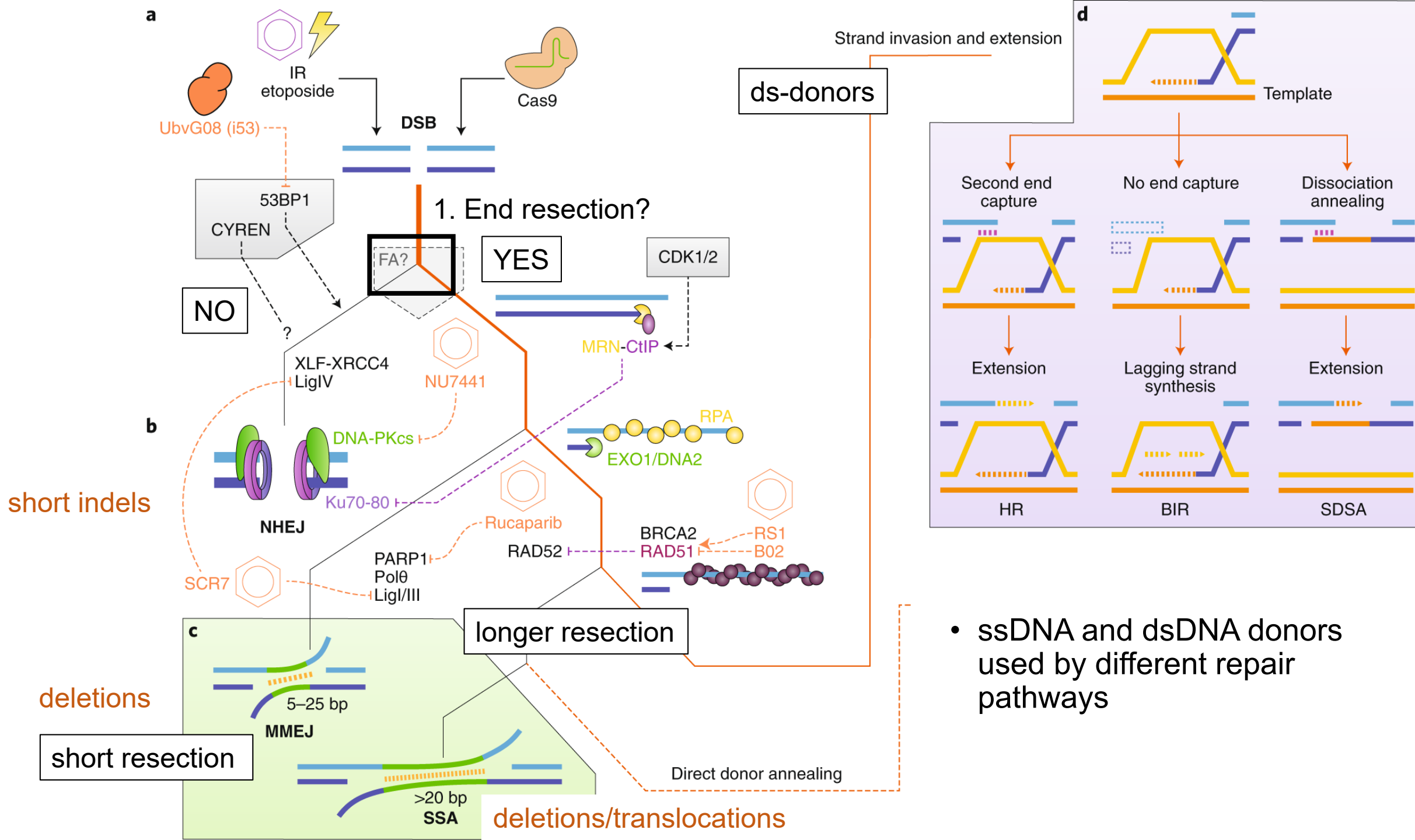


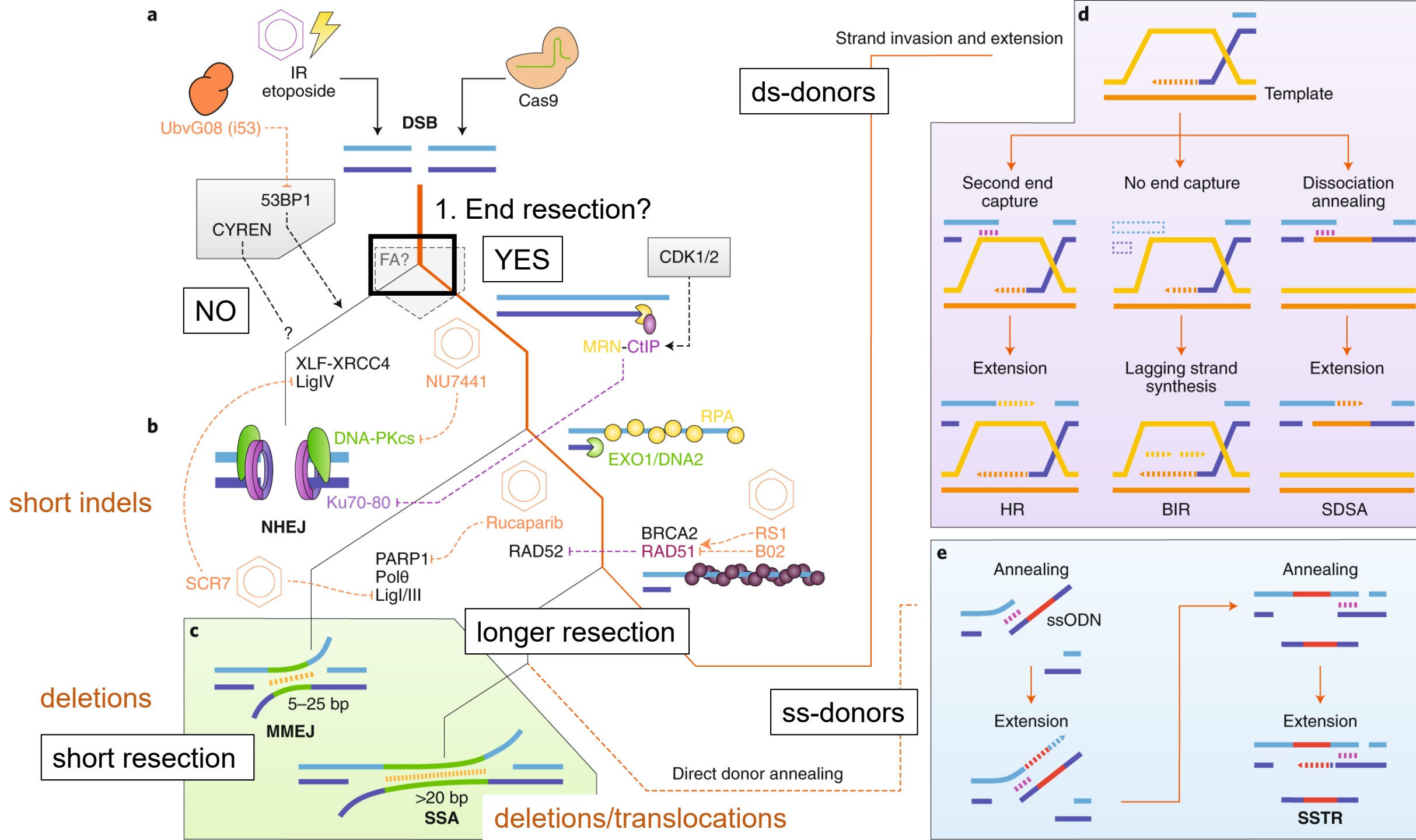


- SSA & MMEJ (Alt-EJ): short homologies around DSB annealed after end resection leading to deletions of variable size or translocations

MMEJ and SSA

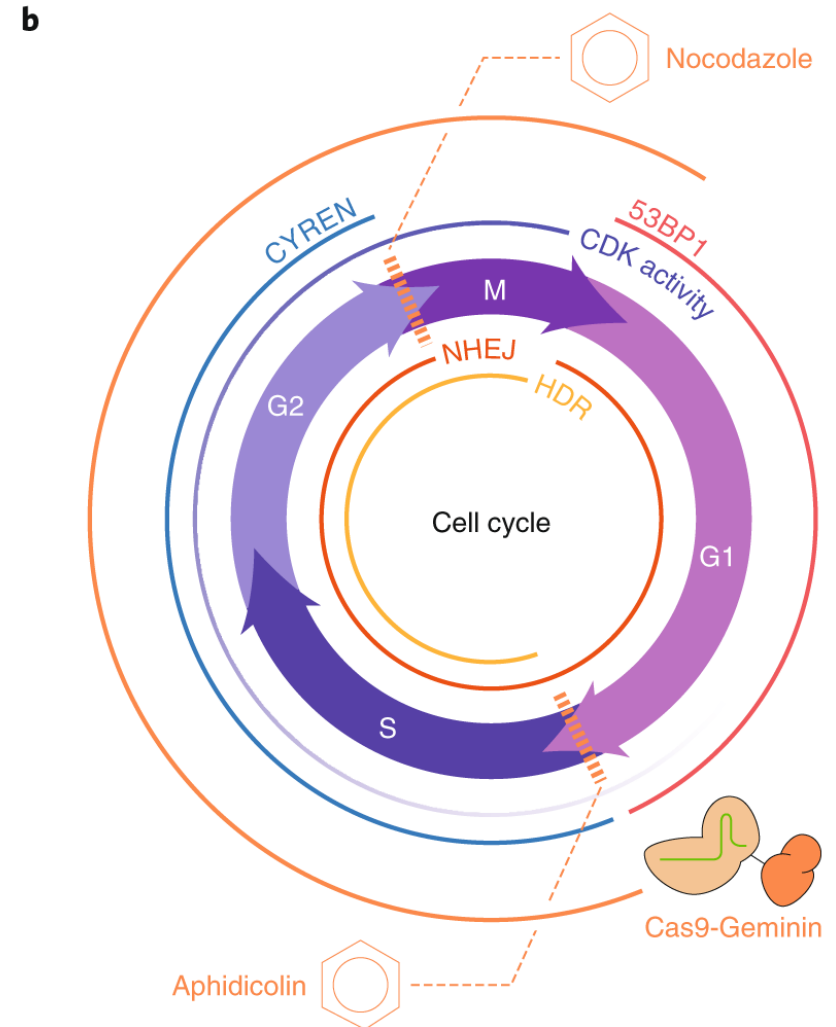






Why is NHEJ (and MMEJ) often more prevalent than HDR?

- NHEJ is active throughout the cell cycle (except M phase) and in many cell types (including postmitotic cells)
- NHEJ is faster (minutes) than HDR (hours)
- NHEJ represses HDR



Alternative repair pathways can be used for precision editing

LETTER

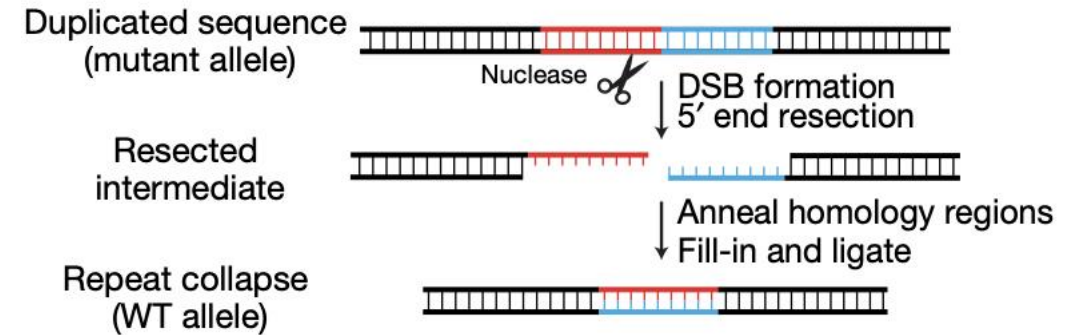
<https://doi.org/10.1038/s41586-019-1076-8>

Precise therapeutic gene correction by a simple nuclease-induced double-stranded break

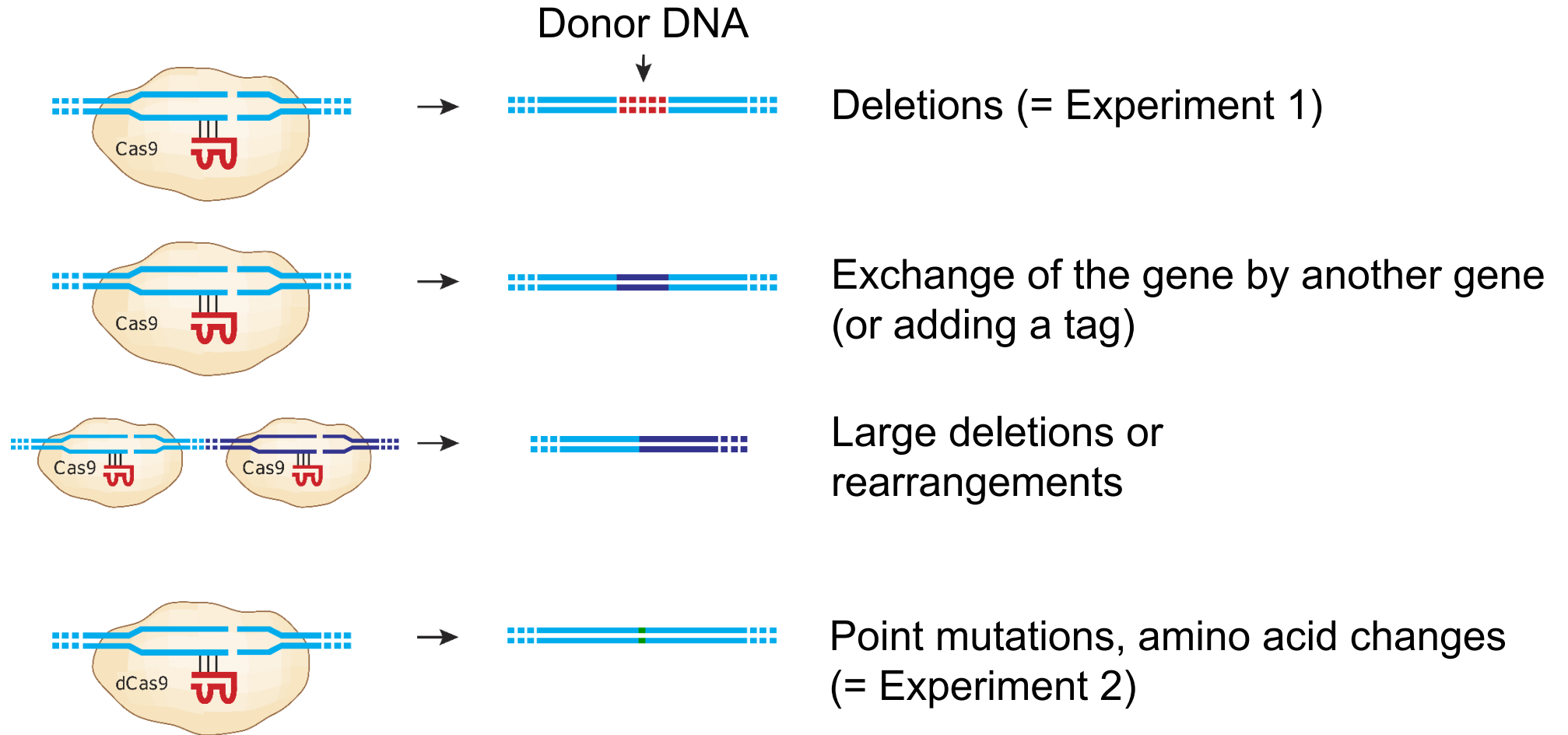
Sukanya Iyer^{1,9}, Sneha Suresh^{1,9}, Dongsheng Guo^{2,3}, Katelyn Daman^{2,3}, Jennifer C. J. Chen^{2,3,7}, Pengpeng Liu¹, Marina Zieger^{4,5}, Kevin Luk¹, Benjamin P. Roscoe^{1,8}, Christian Mueller^{4,5}, Oliver D. King^{2,3}, Charles P. Emerson Jr^{2,3,5*} & Scot A. Wolfe^{1,5,6*}

Limb-girdle muscular dystrophy type 2B

- Weakness and wasting (atrophy) of the muscles of the pelvis and shoulder girdle.
- Most patients eventually need a wheelchair
- No treatment currently



Different types of genome modification via HDR



Through homology-directed repair, different types of modifications can be introduced by designing different repair templates (donor DNA).

gRNA design for HDR

gRNA design considerations for different CRISPR applications

LOCATION
SEQUENCE

Knockout (whole gene deletion, indels)

LOCATION
SEQUENCE

CRISPRi/a

LOCATION
SEQUENCE

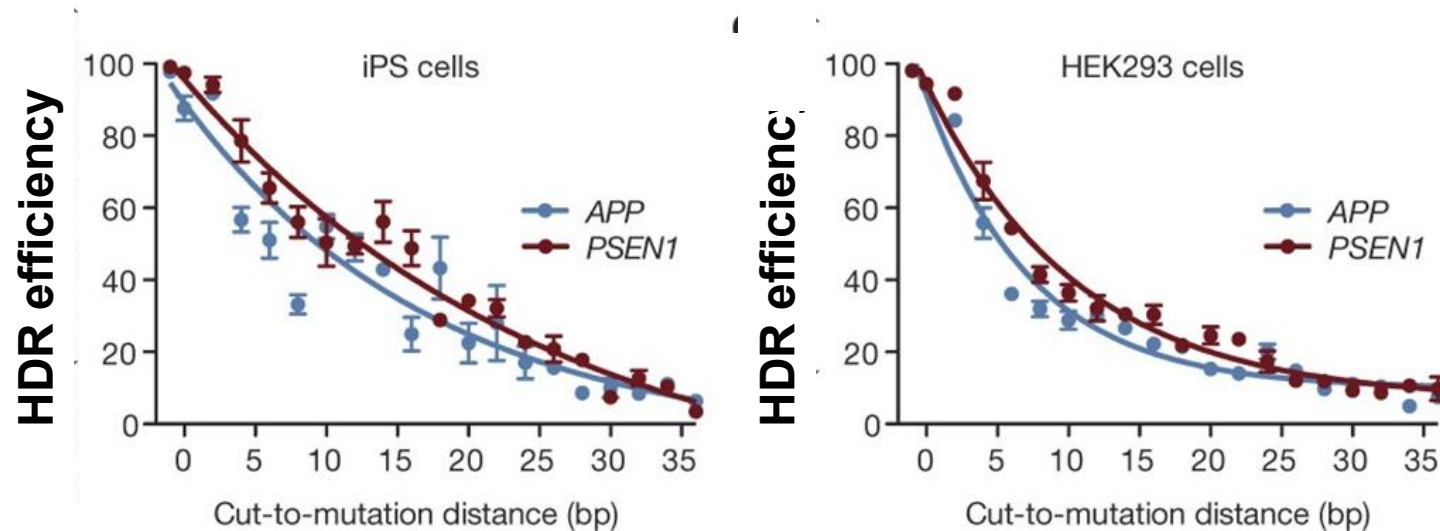
Precision genome editing (base changes)

How to design a sgRNA for HDR editing with *SpCas9*?

- Search for NGG PAMs around your target mutation
- Choose best guide(s) – how?
 - Off-target activity
 - On-target activity
 - Distance of mutation to PAM and protection from recutting



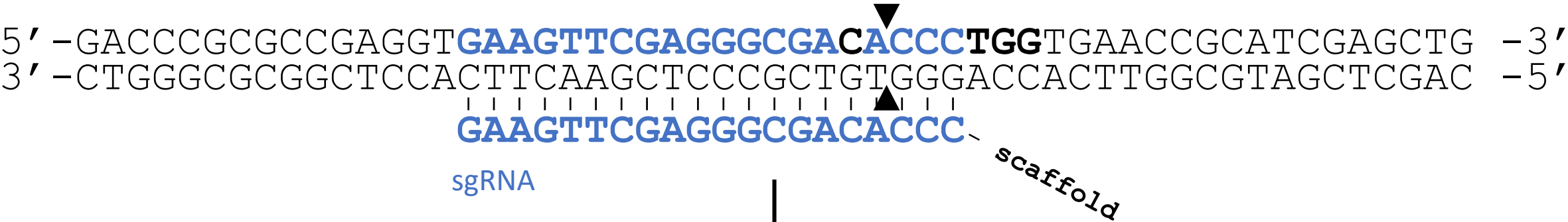
Editing is more efficient for PAM-proximal mutations



If no other mutation introduced:

- mutation in PAM or gRNA region to prevent re-cleavage & donor cleavage

Pre-edit



C to G SNP introduced via donor 5 bp upstream of PAM

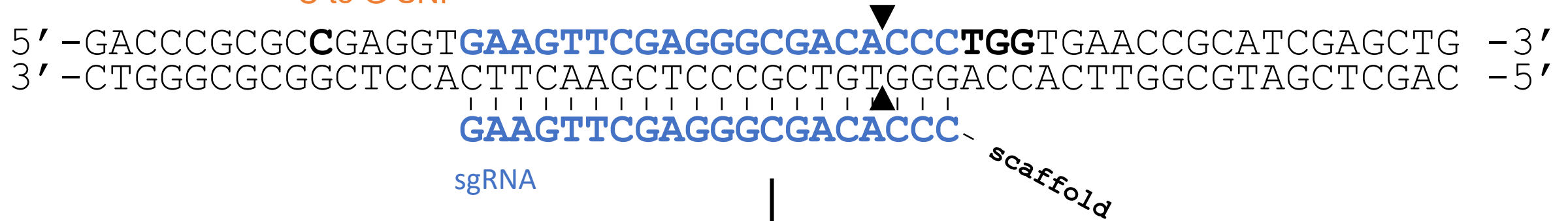
Post-edit



C-C mismatch → target no longer cleaved by Cas9 RNP

Pre-edit

C to G SNP



C to G SNP introduced via donor 26 bp upstream of PAM

Post-edit

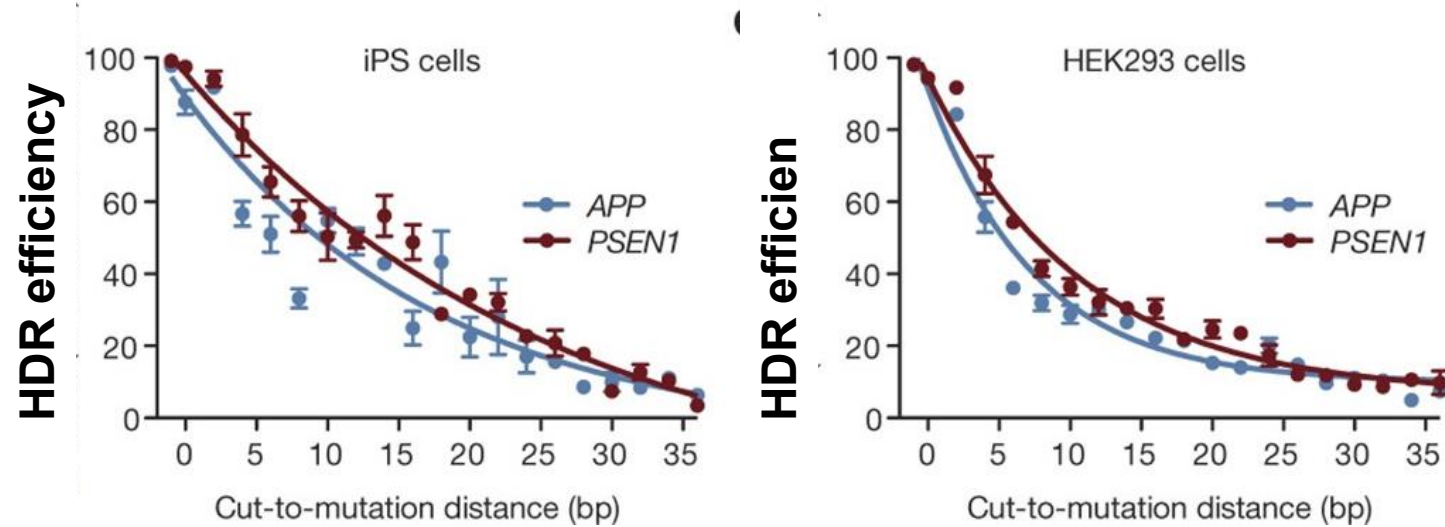
C to G SNP



Perfect match → target cleaved by Cas9 RNP

Same logic applies to donor DNA!

Editing is more efficient for PAM-proximal mutations

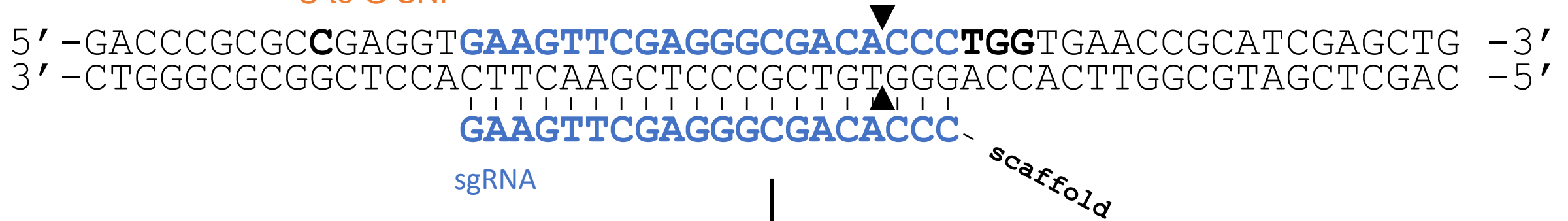


If no other mutation introduced:

- mutation in PAM or gRNA region to prevent re-cleavage & donor cleavage
- ideally PAM or up to +12 nt from PAM
- Blocking mutations in donor to extend range, BUT no single mutations anymore

Pre-edit

C to G SNP



C to G SNP introduced via donor 26 bp upstream of PAM
and blocking mutation in PAM (TGG to TTG)

Post-edit

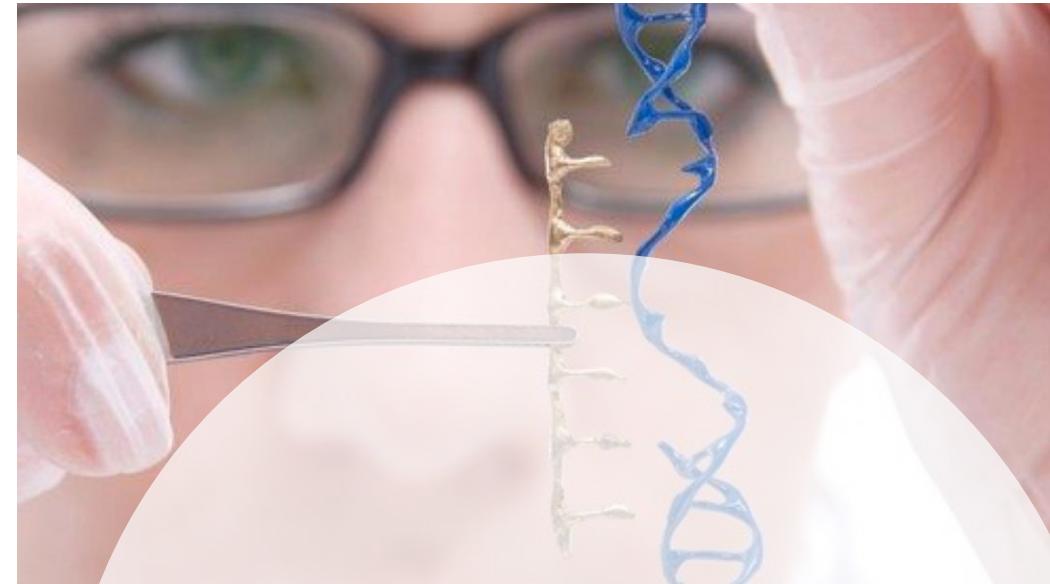
C to G SNP



No PAM motif → target no longer cleaved by Cas9 RNP

Task: Design good gRNAs for your two assigned mutations

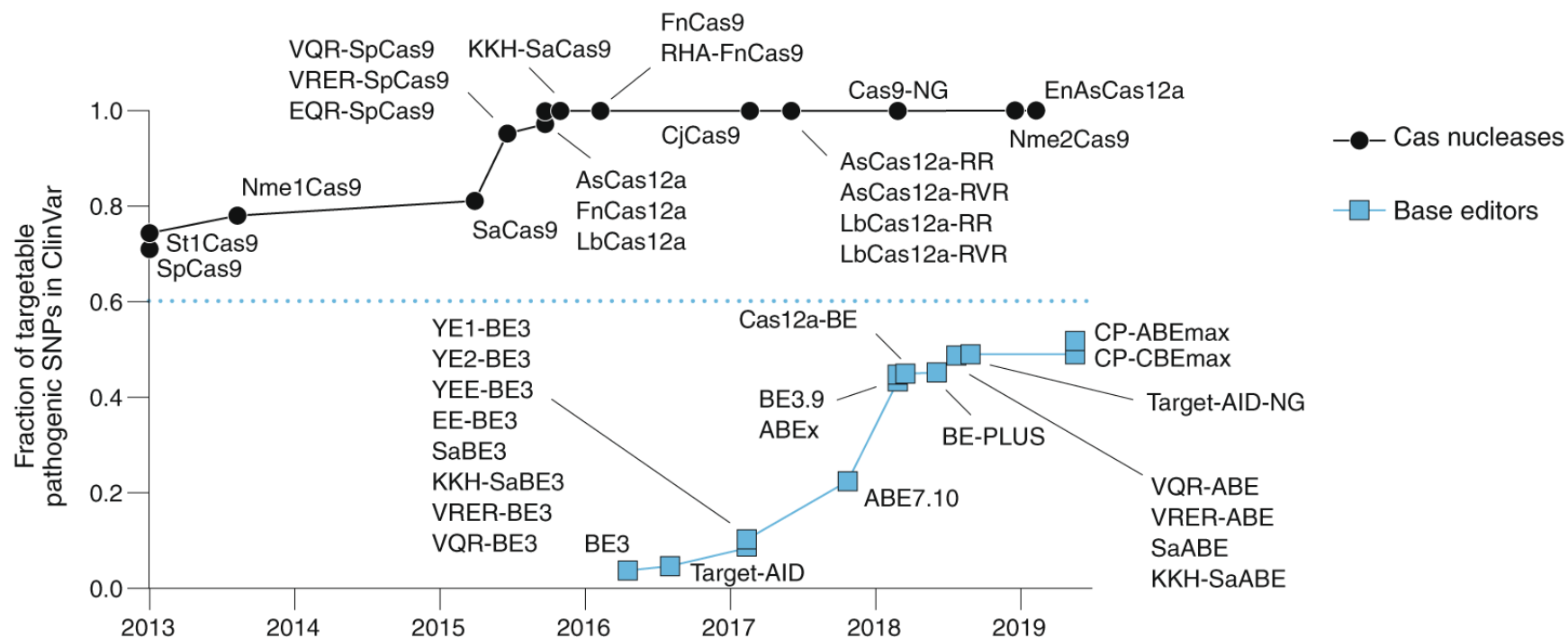
- Visualize possible gRNAs on sequence, annotate with efficiency/off-target predictions, PAM distance
- Select gRNAs for each mutation that you think would work well. Motivate your choice (why this one and not another).



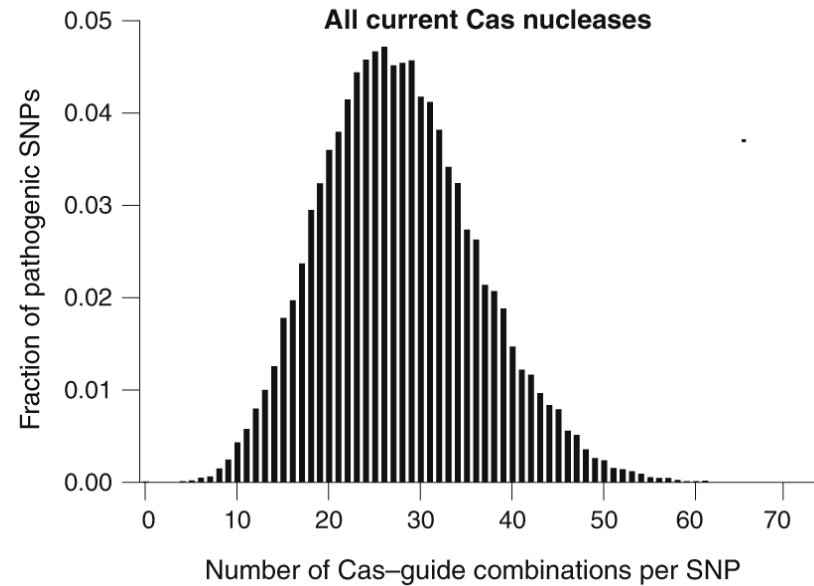
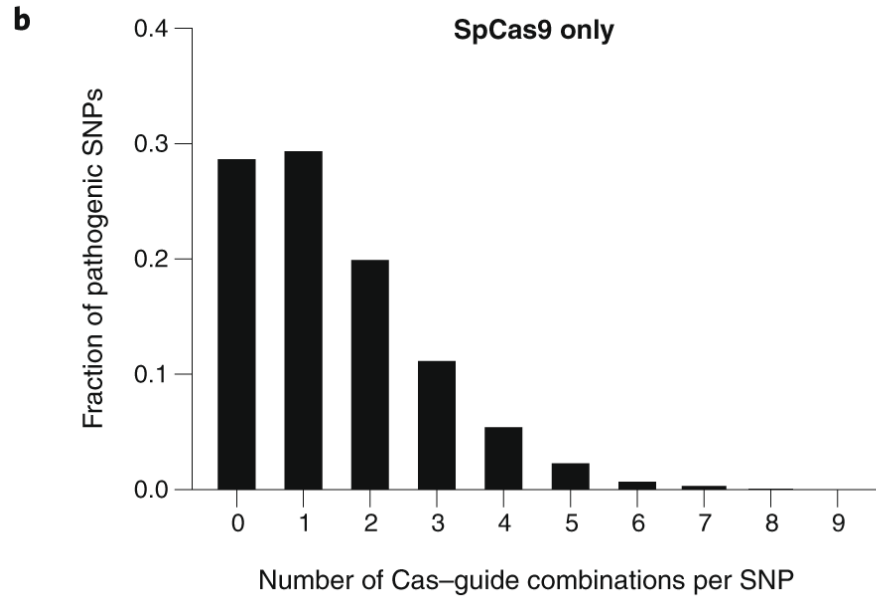
gRNA design
session 2

How much of the genome can we target?

a

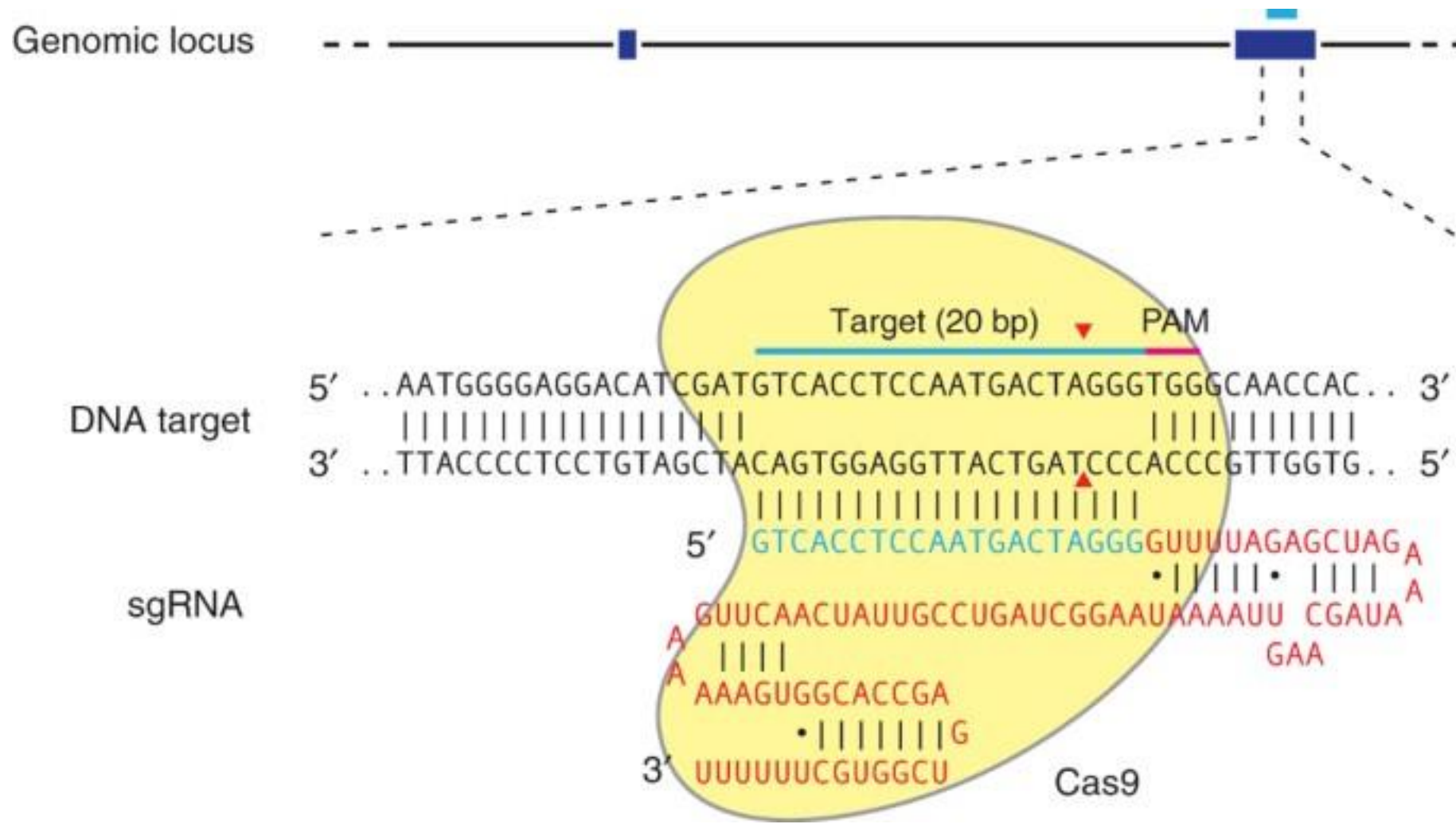


How much of the genome can we target?



- Cas9-VQR (NGA PAM): [10.1038/nature14592](https://doi.org/10.1038/nature14592)
- Cas9-NG (NG PAM): [10.1126/science.aas9129](https://doi.org/10.1126/science.aas9129)
- SpRY-Cas9 (NRN, NYN PAM): <https://doi.org/10.1126/science.aba8853>
- Cas12a nucleases (T-rich PAM)

Donor design



Donor structure

Target locus



Donor



Deletions: HAs flank region to be deleted
Knock-in: HAs flank insertion site

Donor design considerations

- Plasmid or linear?
- Single- or double-stranded?
- Symmetry?
- Homology arm length?

→ Often multiple possibilities, most efficient format can vary depending on organism and type of modification

Donor design general guidelines

- In most organisms, single or double-stranded donors, linear or plasmid encoded work fine → whatever is easiest/cheapest.
- Typically symmetric around mutation (left and right homology arms (HAs) are the same length), but in some cases asymmetric donors can boost efficacy.
- Knock-in needs longer HAs (500-1000bp) than SNPs (40-80 bp)
- Some organisms need longer HAs even for SNPs: if you don't know, try both

Donor design for the course (*E. coli*)

- Single stranded linear donors (lagging strand*) or double-stranded donors (linear or plasmid-encoded)
 - course: linear dsDNA
- Symmetric or asymmetric works well
 - Course: symmetric
- 40 - 50 bp HA for SNPs & deletions, longer HAs beneficial for several mutations, knock-in

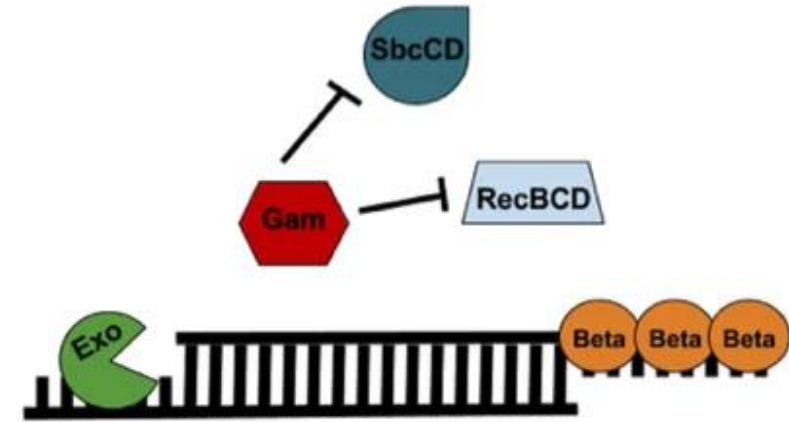
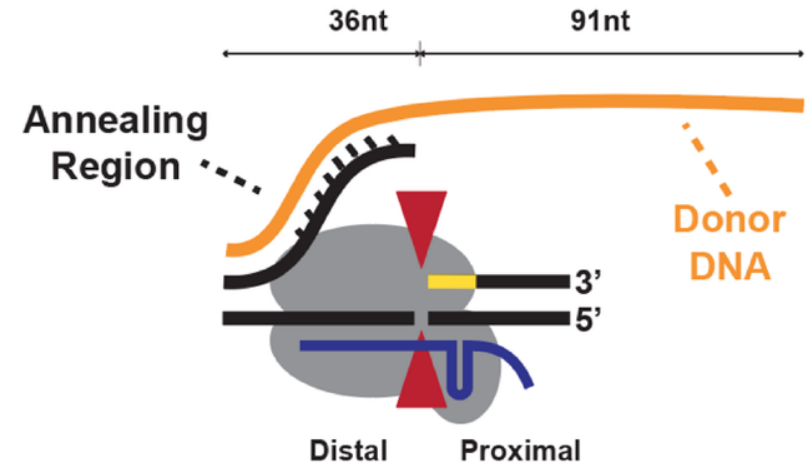
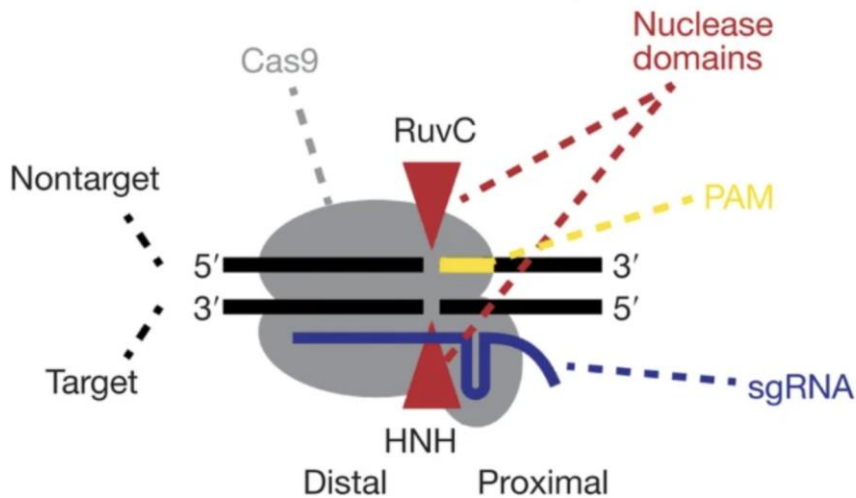


Figure from Addgene blog

*Leading or lagging? https://link.springer.com/protocol/10.1007/978-1-0716-2233-9_9#Sec14

Donor design mammalian (animal) cells – SNPs, small inserts

- ssODN (single stranded DNA) - lowest toxicity
 - 30-90 nt per HA
 - Asymmetric for best efficiency
 - Target (gRNA complementary) or nontarget strand?
Depends on cell type, locus, ...



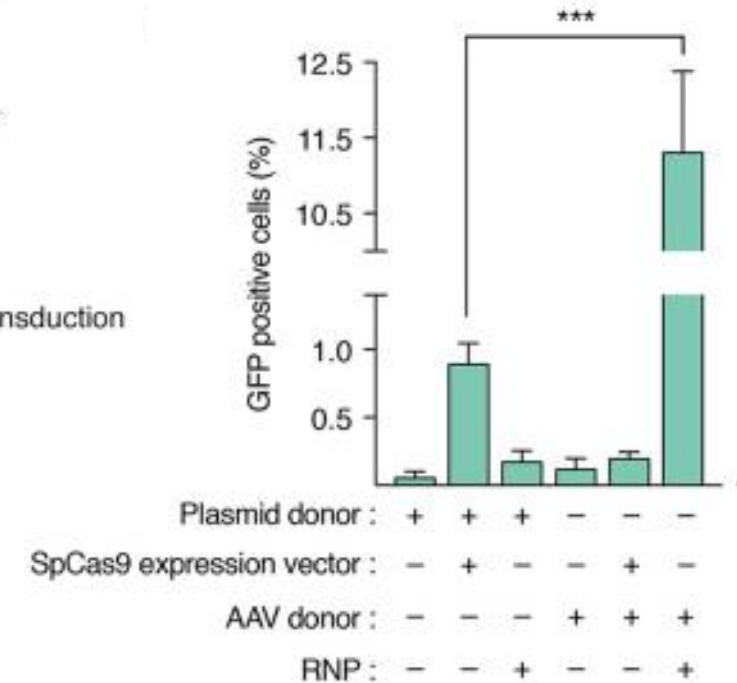
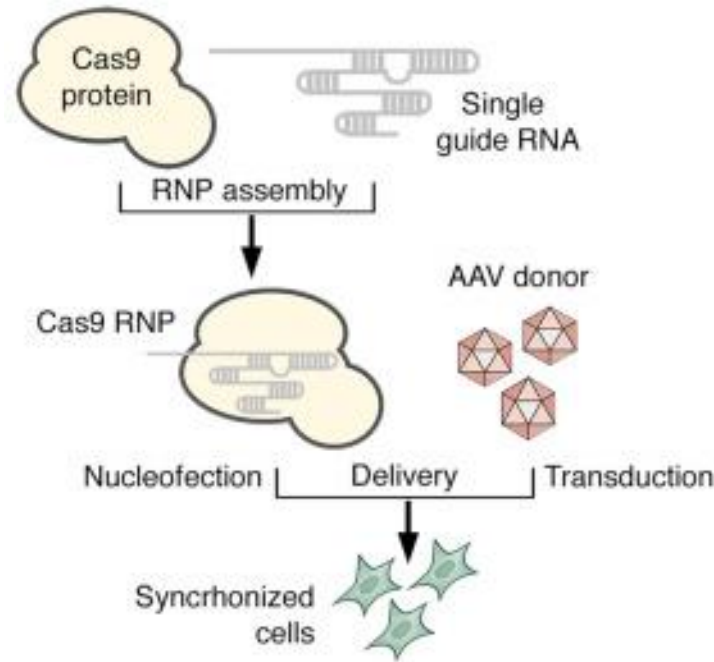
<https://www.nature.com/articles/nbt.3481>

- Blocking mutations may enhance efficiency (i.e. 2 mismatches to target)
- ssODN stability improved via phosphothiorate modifications of last 2 nucleotides

See also: <https://www.nature.com/articles/s41598-021-98965-y>: what if my mutation is outside gRNA region, donor design for Cas12a

Donor design mammalian (animal) cells – larger inserts (>100-200 bp)

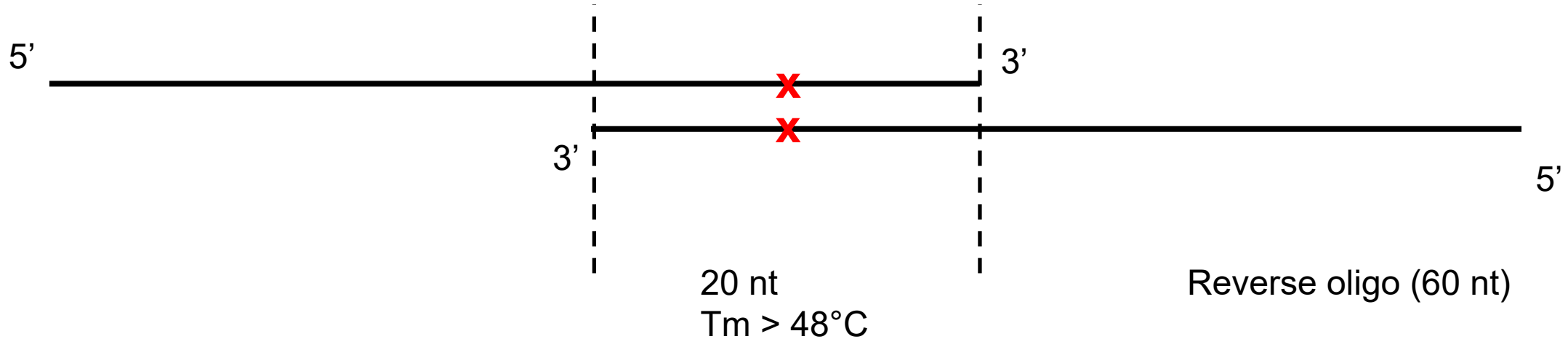
- ssDNA or AAV donor for <1.5 kb inserts
 - Generate ssDNA for larger inserts: Easi-CRISPR ([10.1186/s13059-017-1220-4](https://doi.org/10.1186/s13059-017-1220-4)): *in vitro* transcription and RT
- dsDNA plasmid or AAV donor for larger inserts
- 500 – 1000+ bp HAs plasmid donors (optionally linearized)
- 200 – 800bp AAV donor vector
- Larger inserts – longer HAs



Task: design donor oligos for your assigned PTC

Method: 1-cycle extension of overlapping oligos

Forward oligo (60 nt)

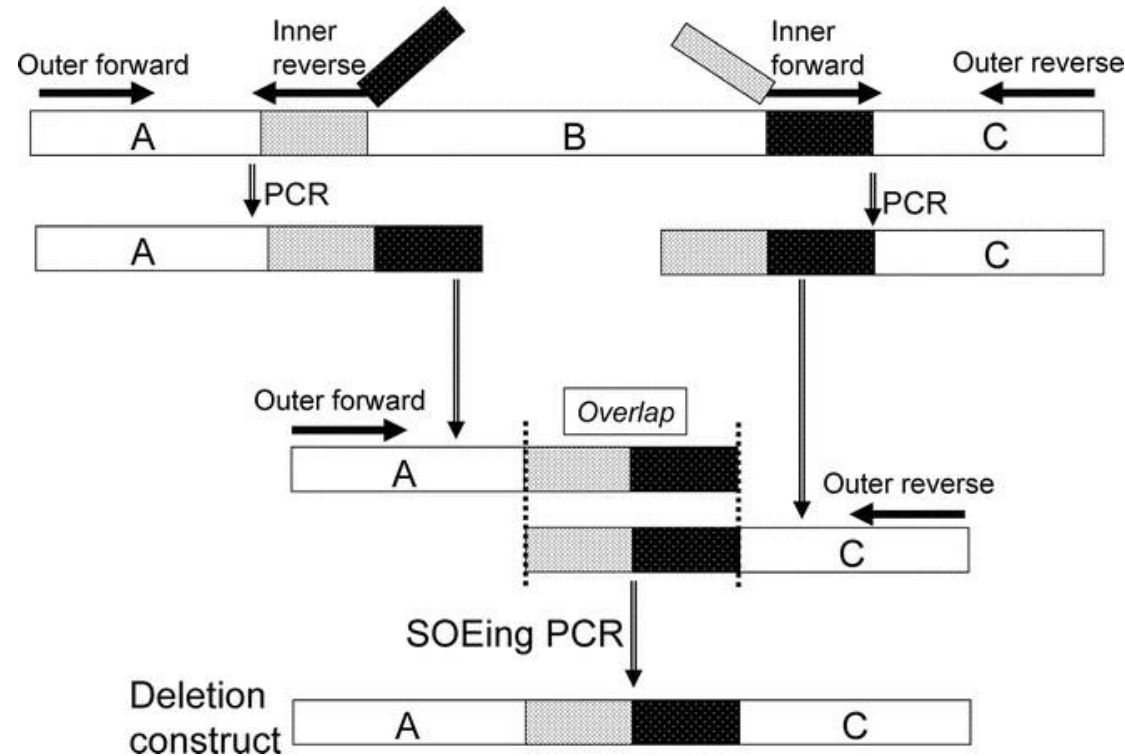


For our experiment: design primers to generate ca. 50 bp homology arms

How to design donor oligos for a *galK* whole-gene deletion

Method: SOEing

B = gene to delete



See separate protocol

Selected gRNAs and donors:

<https://benchling.com/s/seq-cl9LHXPhGCVQVnWkY9s5?m=slm-A0QDDaillwrmUXFBKYr4>

Group 1 & 4 – gRNA 1 & donor 1 (PTC), gRNA 1 + WGKO up, WGKO dwn (whole gene deletion)

Group 2 & 5 – gRNA 2 & donor 2 (PTC), gRNA 2 + WGKO up, WGKO dwn (whole gene deletion)

Group 3 & 6 – gRNA 3 & donor 3 (PTC), gRNA 3 + WGKO up, WGKO dwn (whole gene deletion)

Annotate gRNAs with predicted efficacy, off-targets in your notebook